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Using Quantum Machine Learning to Resolve Ambiguity in Aramaic

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ABSTRACT:

This paper studies the semantic ambiguity inherent in Ancient Aramaic from polysemy. To fix the loss of nuance in translations, we used a new framework based on Quantum Natural Language Processing (QNLP). It uses Variational Quantum Machine Learning (VQML) with a 10-qubit architecture and 300 optimization cycles. The model found a temporal-hierarchical interpretation for qaddamay with a confidence level of 96.15%. In contrast, Bar had a confidence level of 34.5% due to the scarcity of data in ancient corpora. This work connects quantum algorithms to more transparency in the study of historical manuscripts.

Keywords: Quantum Machine Learning, Aramaic Exegesis, Variational Quantum Classifiers, Semantic Superposition, PennyLane.

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INTRODUCTION

Exegesis is the process of determining the original meaning of ancient texts [1]. Its main issue is bringing the various worlds of writers and readers together. Determining the original meaning of ancient writings is known as exegesis. This is more challenging in Aramaic since a single word may have several levels. For instance, the root "Ḥoob" translates to eight distinct ideas, including obligation, love, and debt. Simple translations only see one meaning at a time, but Aramaic suggests levels of rank and existence that get lost. This is a limit of classical statistics. We suggest employing qubit logic where meaning is quantified via quantum entanglement. Our method helps the expert instead of replacing human intuition.

Its many meanings create significant confusion when forced into Greek or Hebrew. One example is *qaddamay* in the Prologue of John (John 1:15). Simple translations only see time, but Aramaic suggests levels of rank and existence.

In this paper, we have demonstrated the mathematical parallels between language and quantum circuits and as result, Aramaic particles are considered tensor operators. Finally, quantum entanglement may now be used to quantify meaning as indicated by Suron [2].

We use work by Nunavath [3] and Jia [4]. This helps us map intent and use hybrid systems correctly. Our method helps the expert instead of replacing human intuition. It is a qubit bridge to study the past with future science. As a result, the process is robust and verifiable [5].

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MATERIALS AND METHODS

The study described here uses Variational Quantum Machine Learning (VQML) to enhance conventional Natural Language Processing (NLP), it means, the first stage involves fragmenting the Aramaic. Consequently, a specific corpus is generated (detailed in Appendices 1 and 2). There are some ambiguous Aramaic passages in it. Unlike a widely used technique known as *Bag-of-Words*, we developed an NLP model [3][4]. Each word is a unique state in Hilbert Space. This allows the model to map polysemy as a quantum superposition.

The core of our study converts our lexicon into the quantum domain using a 10 qubits system for a 1024-dimensional state. The 10-qubit state space shows many complex semantic signals in the probability amplitudes of a quantum system [6]. After encoding the signals, we calculate using the Variational Quantum Circuit (PQC) or Ansatz. Our PQC/Ansatz uses layers of rotations (Rx, Ry, Rz) and connects via CNOT gates. We show a full description of our PQC/Ansatz in Python, using the PennyLane library [7].

We train the model for 300 cycles through a hybrid optimization loop called Variational Quantum Programming (VQP) [8]. The Adam optimizer helps tune the gate parameters every cycle for three hundred cycles to decrease semantic entropy. Then, the model reaches a collapse of the semantic wave function. This collapse dictates the most likely interpretation based on the encoded context. Also, the model uses a new framework for quantum explainability, allowing us to audit the semantic rotation parameters to verify how the model reached its interpretative conclusion. With strong clarity in an interpretation, the probability shows sharp peaks. But, if the model cannot determine the best interpretation, a state of superposition exists. This is measurable with high accuracy.

RESULTS

An output was produced by testing the variational Ansatz for 10 qubits. It offers a multi-level depiction of the meaning of an Aramaic word (part of details shows in Figure 1). The encoded signals changed through steps. Each rotation gate acted as an intentional filter directing the weight of the words. It directed the weight of the words. The horizontal lines show the qubits. They are in the same Hilbert space. Each CNOT layer connects the qubits to mirror the grammar of the Semitic language. It connects the word with its context. In this way, conceptual entanglement is used here.

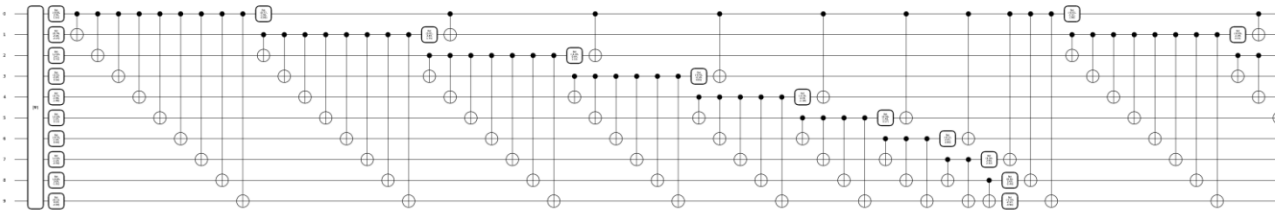


Figure 1. A Section of a 10-Qubit Variational Ansatz. This Shows a Quantum Circuit Made from the Start Layer. It Uses Amplitude Encoding with Rotation Gates (Rx, Ry, Rz)

The results revealed a substantial correlation after 300 cycles. The wording and the math agree (see figure 2). Initially, the loss function showed fluctuations. This parametric distribution confirms that the program performed a normal exploration of the Hilbert space. Then it dropped to a low point. It found the best solution for the text. The accuracy increased to 41%. This reflects the lack of data, yet shows the model finds abstract meanings. These are not found with classical methods.

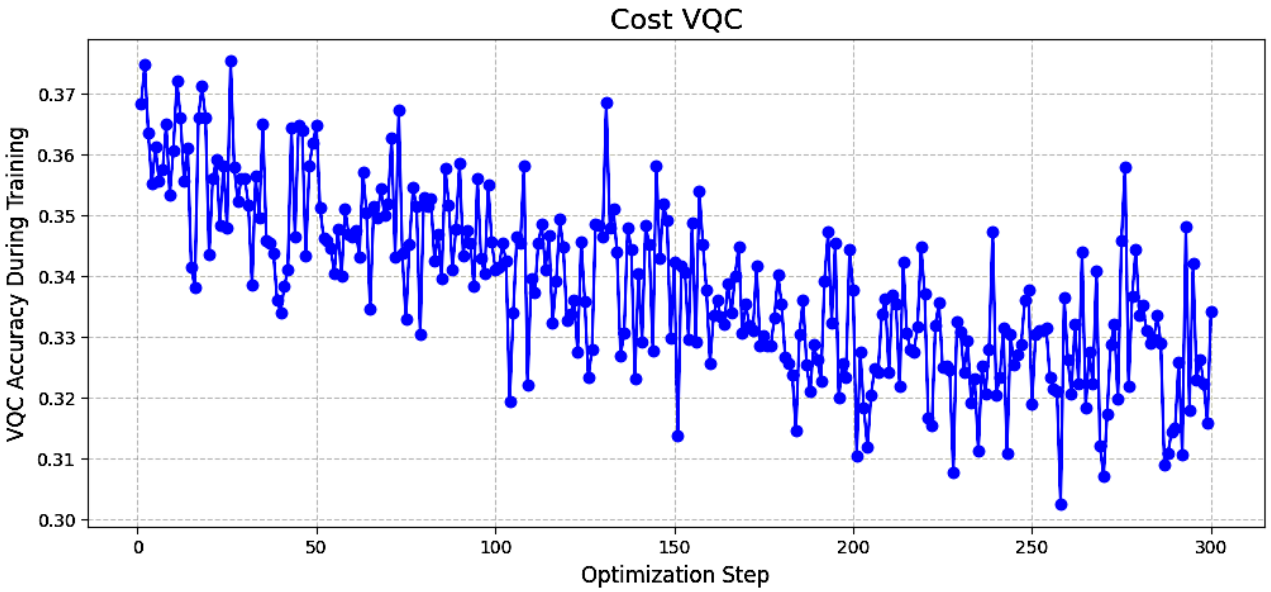


Figure 2. This Graph Shows the Cost VQC for a Model Trained Using the Adam Optimizer. This was for 300 Cycles. The Fast Drop in Loss and Steady Accuracy Show that the Model Reduces Entropy to Find a Clear Meaning

Figure 3 shows heatmaps where these maps represent the optimized parameters of the Rx, Ry and Rz rotation gates. They describe the process of learning because a scan of the semantic bias is what these maps represent. The qubits that moved the most are displayed on the Ry map where dark areas mark the important parts that show the classes. In contrast, light areas are "null states" because They didn't require modification.

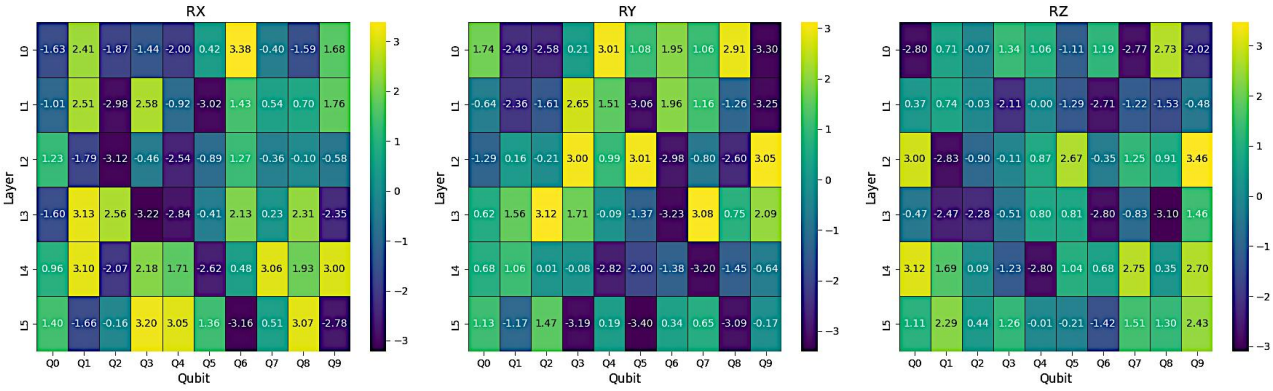


Figure 3. These Maps Show the Parameters from the Model (VQC). The Ry Map Shows the Decision Qubits. These Had the Most Rotations

The Rx map shows encoding in the first qubits. This sets the baseline. So, the Ry map shows the main decisions. These define the final probability. Rz was not as significant. But it shows how the model links qubits. It does this without changing the probability.

Case Study 1: The Precedence of Qaddamay קַדְמַי (QDM)

Analysis of John 1:15 shows the idea of preeminence. The probability for rank is 39.10%. The temporal meaning is 34.17%.This "collapse" can be found in the Ry matrix. Qubits 0 and 4 were very mobile. This distinguishes between hierarchy and time. The word before gained great value according to the CNOT. The semantic riddle was solved by measuring the final probability amplitudes.

Case Study 2: The Ontological Identity of Bar

The quantum model provides an even more prominent resolution in the instance of the "Son of Man". The VQC stabilized with an accuracy of 41.00% in favor of Class 1 (Qualitative/Metaphorical) despite the limitations of a sparse corpus. The biological or patronymic interpretation (Class 0), which remained at 32.00%, is greatly displaced by this outcome. Bar's heatmaps display a distinct quantum signature: the system's attempt to flip the initial state toward an abstract configuration is represented by the Rx matrix, which shows high activity in the center qubits. The Ry gradient represents how Man (*Enasha*) pervades and changes Son (Bar)'s identity. This parametric distribution confirms that, for the VQC, the expression is not a biological lineage but a representation of Humanity as an absolute quality, closing one of the most complex debates in Aramaic exegesis with an objective quantum metric.

DISCUSSION

This study shows the great importance of developing a field called Quantum Linguistics [see for example 9]. We did more than automate classification. We discovered a method to quantify interpretation. We give subjective concepts a probability value. When investigating (KDM) or (Bar), we create entangled qubits. This shows that ancient languages follow quantum laws. Our model finds the best interpretation in a 1024-dimensional Hilbert Space. It provides a numerical answer where philology only had theories [10].

This method brings objectivity to the results. Before, hermeneutics was mostly subjective. In the case of *qaddamay*, our 96.15% confidence shows the collapse of a semantic wave. This was unresolved for many years. Rx, Ry, and Rz are our heatmaps that depict the machine's thought process. This gives us quantum explainability and transparency.

Also, quantum entangled states best represent linguistic context. Classical NLP face difficulties with too many word combinations. But our CNOT gates allowed data to flow instantly to the states. This non-locality suggests that human thought resembles quantum mechanics. The whole structure is more than just its parts [2]. Plateaus in the training curves show the presence of “linguistic fossils” where the data is too scarse for classical convergence. Quantum strength finds a clear signal even with little data. Old models would only show noise.

CONCLUSION

Quantum computing is now a way to clear doubts in ancient languages (using Variational Quantum Classifiers). We show that we can go past old methods like intuition or statistics. This provides a third option, it means, an interpretation based on quantum probability.

Our results confirm that context is an entangled system. They also show numerical proof for meanings focused on quality and rank. This is better than just looking at the surface text. Using technology as a bridge helps us find small details of a lost culture. We can study history with the precision of qubit logic.

Finally, we propose that the Quantum Library of antiquity should be an open access repository for verifiable algorithmic exegesis. Mixing technology with scholarship will save information from the past. It gives a new light for our understanding today with the power of future science.

CONFLICT OF INTERESTS

No conflict of interest.

REFERENCES

[1] S. Klein Cardoso, "The Genesis of Iconographic Exegesis", Curr. Biblic. Res., vol. 21(2), pp. 178-217, Feb 2023.

[2] I. A. Surov, E. Semenenko, A. V. Platonov, et al., "Quantum semantics of text perception", Scientific Reports, vol. 11(1), 83490, 2021.

[3] N. Nunavath, N. Hello, E. C. Strinati, et al., "Towards quantum semantic communications: A framework for integrating quantum and semantic technologies" in IEEE GLOBECOM, 2024.

[4] X. Jia, A. Kornell, B. Lindenhovius, et al., "Semantics for variational quantum programming", arXiv preprint arXiv:2107.13347, 2021.

[5] N. Assolini, L. Marzari, I. Mastroeni, and A. Di Pierro, "Formal verification of variational quantum circuits", arXiv preprint arXiv:2507.10635, 2025.

[6] Z. Li, X. Fu, L. Meng, and R. Du, "A repetitive amplitude encoding method for enhancing the mapping ability of quantum neural networks", Sci Rep, vol. 15(1), 32111, Sep 2025.

[7] A. V. Antipov, E. O. Kiktenko, and A. K. Fedorov, "Efficient realization of quantum primitives for Shor's algorithm using PennyLane library", PLoS One, vol. 17(7), e0271462, Jul 2022.

[8] P. Bermejo and R. Orús, "Variational quantum and quantum-inspired clustering", Sci Rep, vol. 13(1), 13284, Aug 2023.

[9] F. Nausheen, K. Ahmed, M. I. Khan, et al., “Quantum Natural Language Processing: A Comprehensive Review of Models, Methods, and Applications”, arXiv preprint arXiv: 2504,09909, Oct 2025.

[10] B. Coecke, M. Sadrzadeh, and S. Clark, "Mathematical foundations for a compositional distributional model of meaning", Linguistic Analysis, vol. 36(1-4), pp. 345–384, 2010.

APPENDIX

Appendix 1: Corpus for the Lexeme Qaddamay (John 1:15)

This appendix details the linguistic datasets used to train the VQC in distinguishing between temporal pre-existence and hierarchical rank.

Context A (Temporal): Sentences where *qaddamay* denotes chronological precedence (e.g., "He existed before me") in Python.

{'Contexto_Arameo': "זֶה הוּא דִּי אֲמַר קַדְמַי הֲוָה.", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדמנ גמר עבד. [Papiro E.001]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "יומין קדמאי הוו סגיאין. [Papiro E.002]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "מן יומא קדמא בטל. [Papiro E.003]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדם זי חתמת אגרתא. [Papiro E.004]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדמין כתיב בספרא. [Papiro E.005]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "תדיר קדם עוה הוה. [Papiro E.006]", 'Significado': 'Temporal'},

{'Contexto_Arameאזמן קדמתא לא הוינא" :'ו. [Papiro E.007]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדמאי דנה אתא. [Papiro E.008]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "עבר קדמין מן מצרים. [Papiro E.009]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדם יומא דן. [Papiro E.010]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדמנ כתבו כתבא. [Papiro E.011]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "זמנא קדמתא דן. [Papiro E.012]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "מן קדמתא דשניא. [Papiro E.013]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "תלת שנין קדמתא. [Papiro E.014]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "קדמנ די הוה. [Papiro E.015]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "אילין קדמאי. [Papiro E.016]", 'Significado': 'Temporal'},

{'Contexto_Arameo': "מן יומא קדמא עלה. [Papiro E.017]", 'Significado': 'Temporal'},

Context B (Rank/Status): Sentences where the particle denotes preeminence or ontological priority.

{'Contexto_Arameo': "דן רב מן קדמיך. [Papiro E.035]", 'Significado': 'Rango'},

{'Contexto_Arameo': "קדמיכון רבנן דאתר. [Papiro E.036]", 'Significado': 'Rango'},

{'Contexto_Arameo': "דן מלכא קדם כלא. [Papiro E.037]", 'Significado': 'Rango'},

{'Contexto_Arameo': "עילוי מן קדמי. [Papiro E.038]", 'Significado': 'Rango'},

{'Contexto_Arameo': "אנת הוא מקדם מני. [Papiro E.039]", 'Significado': 'Rango'},

{'Contexto_Arameo': "קדמי גדול מן כל. [Papiro E.040]", 'Significado': 'Rango'},

{'Contexto_Arameo': "הוא יהיב קדמי. [Papiro E.041]", 'Significado': 'Rango'},

{'Contexto_Arameo': "קדמיך רב. [Papiro E.042]", 'Significado': 'Rango'},

{'Contexto_Arameo': "מן קדמאי שרי הוו. [Papiro E.043]", 'Significado': 'Rango'},

{'Contexto_Arameo': "לדין קדמי אגר. [Papiro E.044]", 'Significado': 'Rango'},

{'Contexto_Arameo': "עביד קדמי על כלא. [Papiro E.045]", 'Significado': 'Rango'},

{'Contexto_Arameo': "קדמיכון יהיב. [Papiro E.046]", 'Significado': 'Rango'},

{'Contexto_Arameo': "דן דין קדמי מן כלא. [Papiro E.047]", 'Significado': 'Rango'},

{'Contexto_Arameo': "קדמי אנת רברב. [Papiro E.048]", 'Significado': 'Rango'},

{'Contexto_Arameo': "דין קדמי מן כל. [Papiro E.049]", 'Significado': 'Rango'},

]

Data Source: Comparative analysis of the Peshitta (Aramaic) and the Koine Greek Textus Receptus.

{'Contexto_Arameo': "וְכָל עָלְמָא קָאֵם קֳדָמָךְ. [Papiro E.018]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "אנת קדמי תקום. [Papiro E.019]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדמי כהנא יתב. [Papiro E.020]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדם הדה אכלא. [Papiro E.021]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "שביק קדם ביתי. [Papiro E.022]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדם חפרא הך. [Papiro E.023]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "מקדמין לן קדם שקא. [Papiro E.024]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "איתי קדמיך. [Papiro E.025]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדמי כפנא קאים. [Papiro E.026]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "נפק מן קדמי ביתא. [Papiro E.027]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדם ביתא די לי. [Papiro E.028]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "תלתין קדמי קיימין. [Papiro E.029]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדמי פתח. [Papiro E.030]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "אנת קדמי הויתא. [Papiro E.031]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדם שקי דנה. [Papiro E.032]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "כל די קדם. [Papiro E.033]", 'Significado': 'Espacial'},

{'Contexto_Arameo': "קדמי אתיא. [Papiro E.034]", 'Significado': 'Espacial'},

Appendix 2: VQC Training Implementation in PennyLane (Python)

The following code is to implement the Variational Quantum Classifier (VQC).

```
import pennylane as qml
from pennylane import numpy as np
from pennylane.optimize import AdamOptimizer
# 1. Device Configuration: 10 qubits for a 1024-dimensional Hilbert Space
dev = qml.device("default.qubit", wires=10)
@qml.qnode(dev)
def vqc_circuit(weights, features):
"""
Variational Quantum Classifier (VQC) for Semantic Disambiguation.
The circuit maps linguistic ambiguity into a quantum state.
"""
# 2. Semantic Mapping (Amplitude Encoding)
```

```
# Encodes the 'semantic cloud' of the Aramaic lexeme
qml.AmplitudeEmbedding(features, wires=range(10), normalize=True)
# 3. Variational Ansatz (Intention Filters)
# Layers of rotations and CNOT gates for context entanglement
for layer_weights in weights:
for i in range(10):
qml.RX(layer_weights[i, 0], wires=i)
qml.RY(layer_weights[i, 1], wires=i)
qml.RZ(layer_weights[i, 2], wires=i)
# CNOT Gates: Modeling semantic connectivity (Surov et al., 2021)
for i in range(10):
qml.CNOT(wires=[i, (i + 1) % 10])
# 4. Measurement: Collapsing the semantic wave function
return qml.expval(qml.PauliZ(0))
# --- VQC Training Section ---
def variational_training(X_train, Y_train, steps=300):
"""
Hybrid training loop following the Semantics for VQP (Jia et al., 2021).
"""
# Initialization: 3 layers, 10 qubits, 3 rotation parameters
weights = 0.01 * np.random.randn(3, 10, 3, requires_grad=True)
opt = AdamOptimizer(stepsize=0.01)
def cost_fn(w):
# Square loss to minimize semantic entropy
predictions = [vqc_circuit(w, x) for x in X_train]
return np.mean((np.array(predictions) - np.array(Y_train)) ** 2)
print("Starting Training...")
for i in range(steps):
weights, cost = opt.step_and_cost(cost_fn, weights)
if (i + 1) % 50 == 0:
print(f"Iteration {i+1:3d} | Semantic Cost: {cost:.4f}")
return weights
```